

1. Executive Summary

The **Online Shopping System** is a web-based application designed to solve common problems found in traditional shopping systems, such as time consumption, limited product availability, and lack of convenience. In manual systems, customers need to physically visit stores, which is inefficient and does not provide real-time product comparison or order tracking.

To overcome these issues, this project introduces an online platform where users can browse products, view details, add items to a cart, and place orders easily from any location. The system also includes an administrative dashboard that allows administrators to manage products, users, and orders efficiently.

The main value proposition of this system is to provide a **fast, user-friendly, and accessible shopping experience**, while also improving business operations through digital product and order management.

Technologies Used:

- Frontend: HTML, CSS, JavaScript
- Backend: Core PHP (no REST API, direct server-side processing)
- Database: MySQL
- Development Tools: XAMPP, VS Code, GitHub

Team Information

Name: ARIF HASAN

ID: 0242220005341044

Role: Backend Developer

Responsibilities: Database design, login system, order processing

Name: YEASIR ARAFAT

ID: 242310005341034

Role: Frontend Developer

Responsibilities: UI design, product display, user interaction

Name: Sayem Islam Tamim

ID: 222-35-1186

Role: Software Testing & Quality Assurance

Responsibilities: Fix bugs , Update & Quality Control

2. Introduction

2.1 Problem Statement

Traditional shopping systems are inefficient because they require physical presence, manual billing, and limited product visibility. Customers cannot easily compare products, track orders, or access services at any time. This creates inconvenience and reduces efficiency.

2.2 Project Objectives

- Develop a responsive and user-friendly shopping platform
- Implement secure user authentication using sessions
- Enable cart functionality and order placement
- Store and manage data efficiently using MySQL
- Provide an admin dashboard for full system control

2.3 System Boundaries (Scope)

✓ System Includes:

- User registration and login
- Product browsing and search
- Add to cart and checkout system
- Order storage and retrieval
- Admin product management

✗ System Excludes:

- Real-time online payment gateway (only simulated)
- Physical delivery handling
- Mobile application version

2.4 Stakeholder Analysis

Stakeholder	Role	Needs
Customer	End user	Easy shopping, fast checkout
Admin	System manager	Full control over products & users
Seller	Product provider	Add/update products
Delivery Agent	Logistics	Update delivery status



3. Requirement Specification

3.1 Functional Requirements

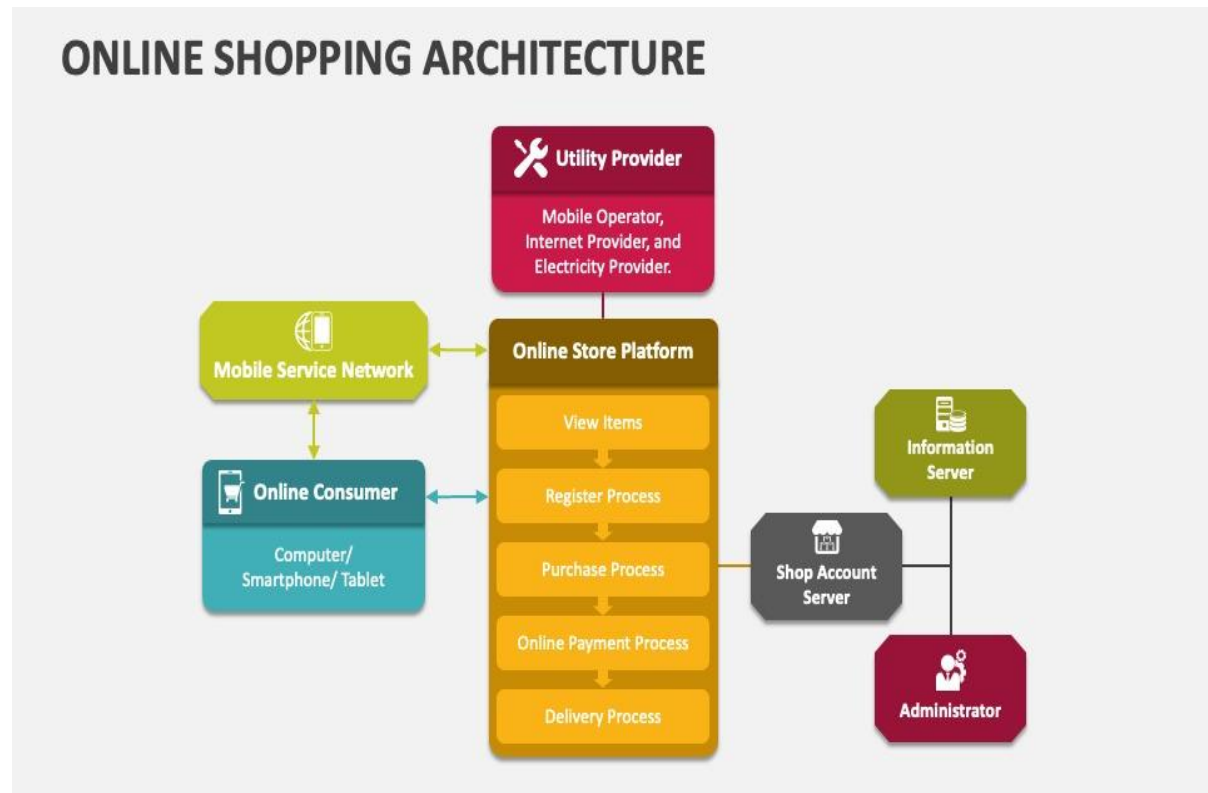
ID	Feature	Description	Priority	Status
FR-1	Registration	Create account	High	Implemented
FR-2	Login	Authenticate user	High	Implemented
FR-3	Product Listing	Display products	High	Implemented
FR-4	Product Details	Show full info	Medium	Implemented
FR-5	Add to Cart	Store items in cart	High	Implemented
FR-6	Cart Management	Update/remove items	Medium	Implemented
FR-7	Checkout	Confirm order	High	Implemented
FR-8	Order History	View orders	Medium	Partial
FR-9	Admin Panel	Manage system	High	Implemented

3.2 Non-Functional Requirements

- **Security:** Session-based authentication and input validation
- **Performance:** Page load under 2 seconds
- **Scalability:** Supports multiple concurrent users
- **Usability:** Simple UI for all user levels
- **Reliability:** System uptime above 95%

4. System Design & Architecture

4.1 Architecture Diagram

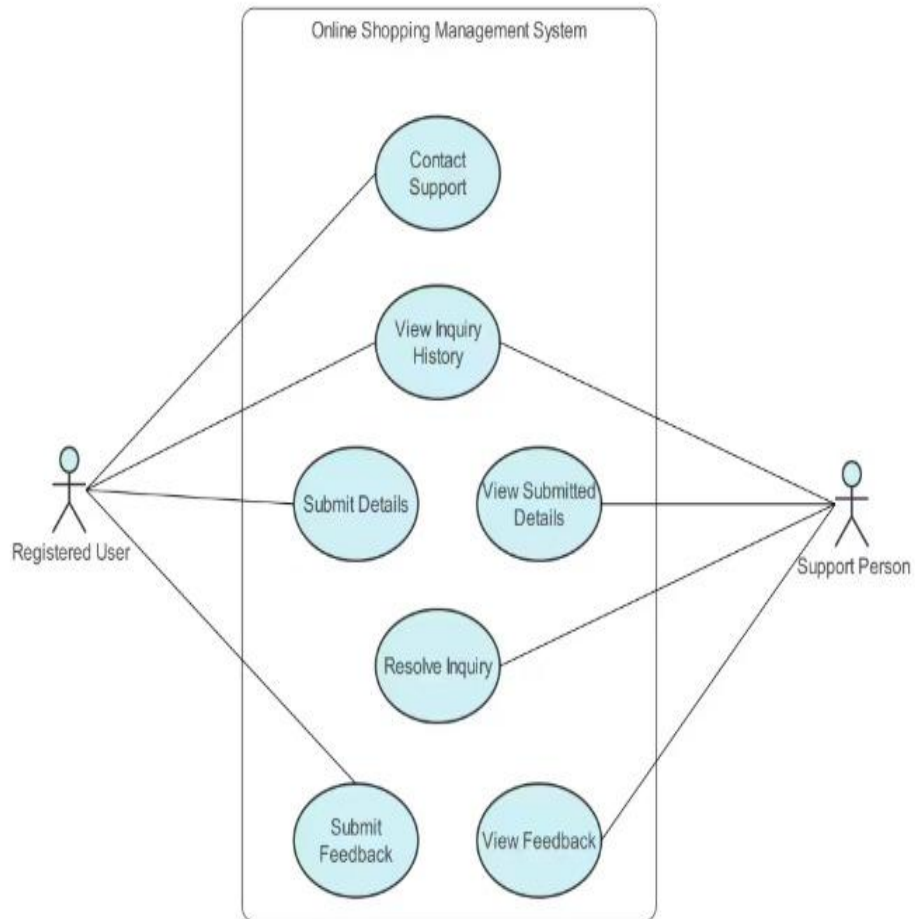


Data flow:

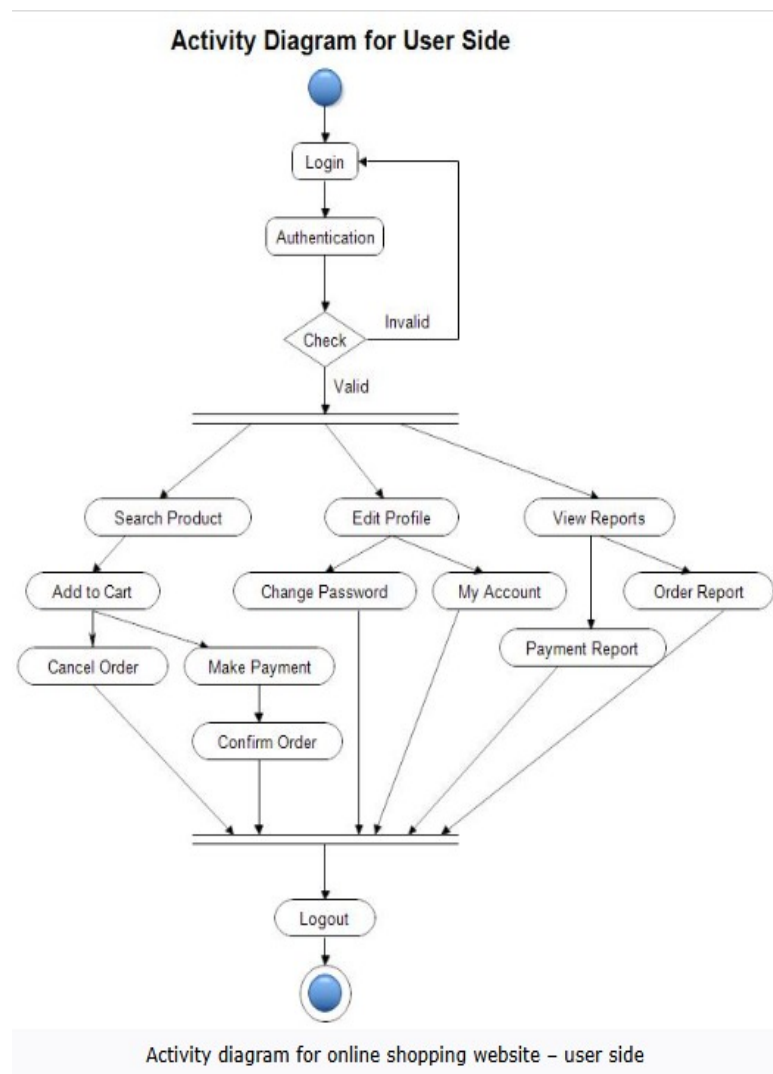
User → Request → PHP → Database → Response → User

4.2 System Modeling

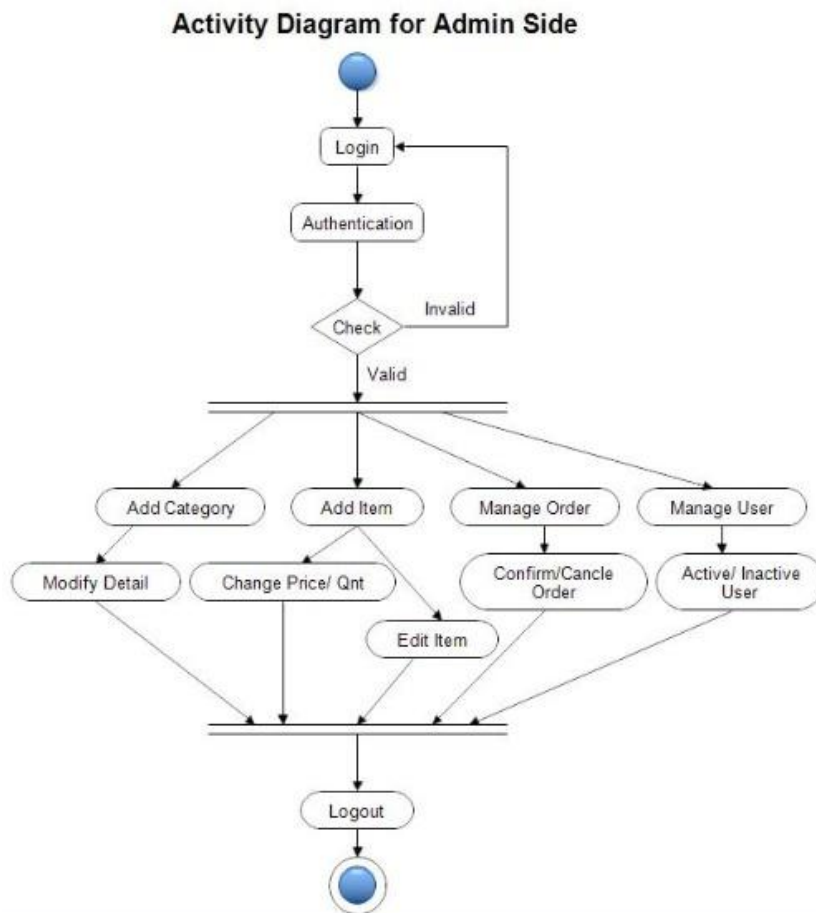
Use Case Diagram



✓ Activity Diagram (User Side)

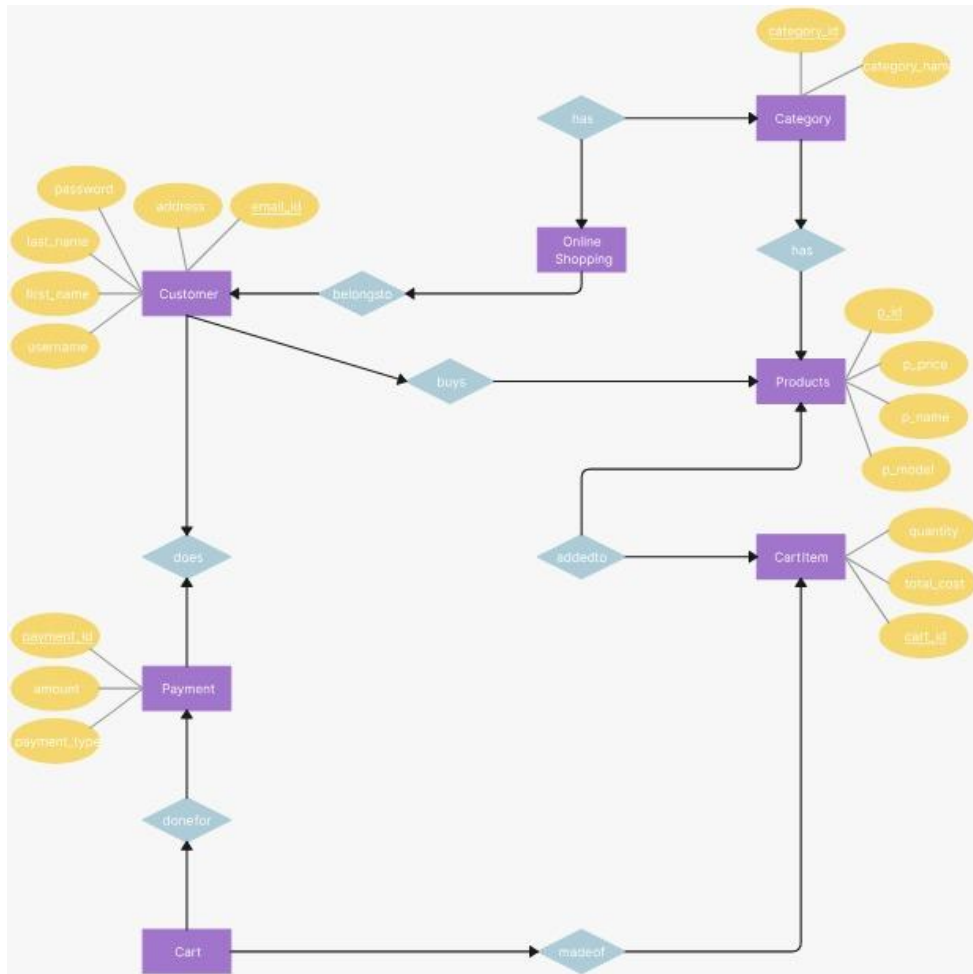


✓ Activity Diagram (Admin Side)



Activity diagram for online shopping website – admin side

4.3 Database Design



✓ Data Dictionary

Field	Type	Constraint	Description
id	INT	PK	Unique ID
name	VARCHAR	NOT NULL	Name
email	VARCHAR	UNIQUE	User email
password	VARCHAR	NOT NULL	User Password
price	INT	NOT NULL	Product price

5. Technical Implementation

5.1 Data Handling (Advanced Explanation)

Process	File	Logic
Registration	register.php	INSERT query
Login	login.php	SELECT query
Product Show	index.php	SELECT all
Cart	cart.php	Session storage
Order	payment.php	INSERT loop

Important:

- No API used
- Direct PHP → MySQL interaction

5.2 RBAC

System checks user role using session:

```
if($_SESSION['role'] == 'admin'){  
    // admin dashboard  
}  
else{  
    // normal user  
}
```

Admin can:

- Add/edit/delete products

User can:

- Only browse and order

5.3 Frontend Organization

Folder structure:

/admin

/css

/js

/pages

Reusable components:

- header.php
- footer.php

Code reuse improves maintainability

6. User Manual (Step-by-Step)

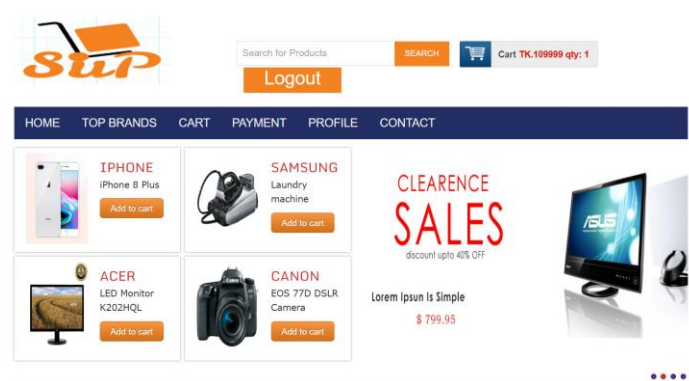
6.1 Getting Started

1. Open website
2. Register account
3. Login

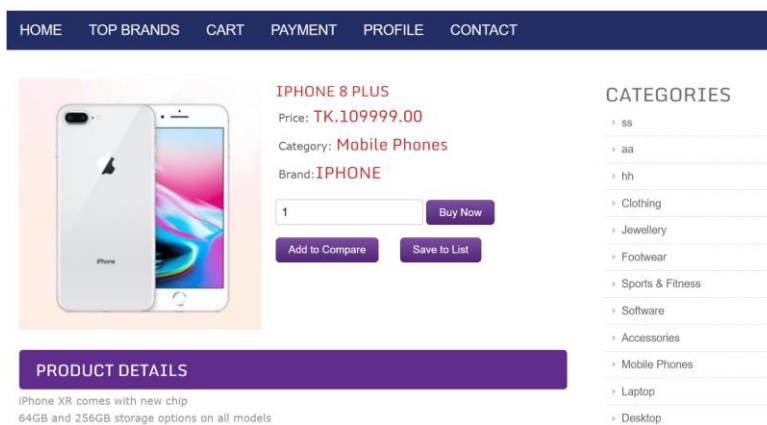
6.2 Feature Walkthrough

✓ Product Purchase

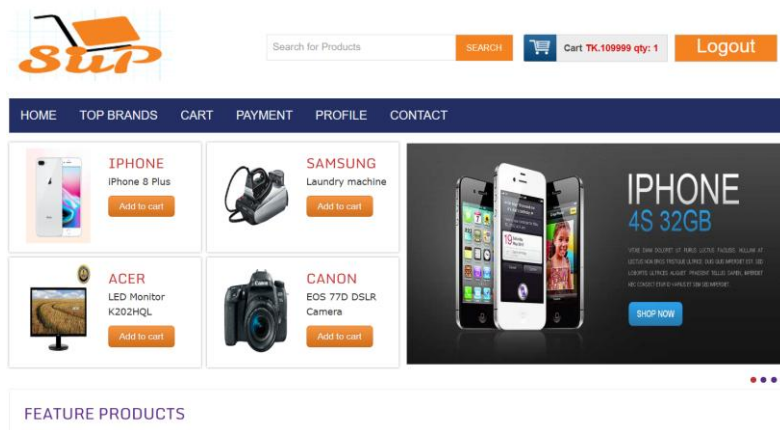
1. Open homepage



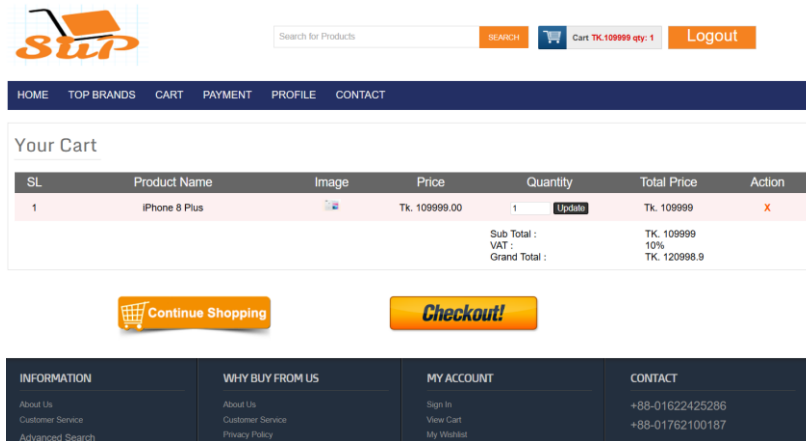
2. Select product



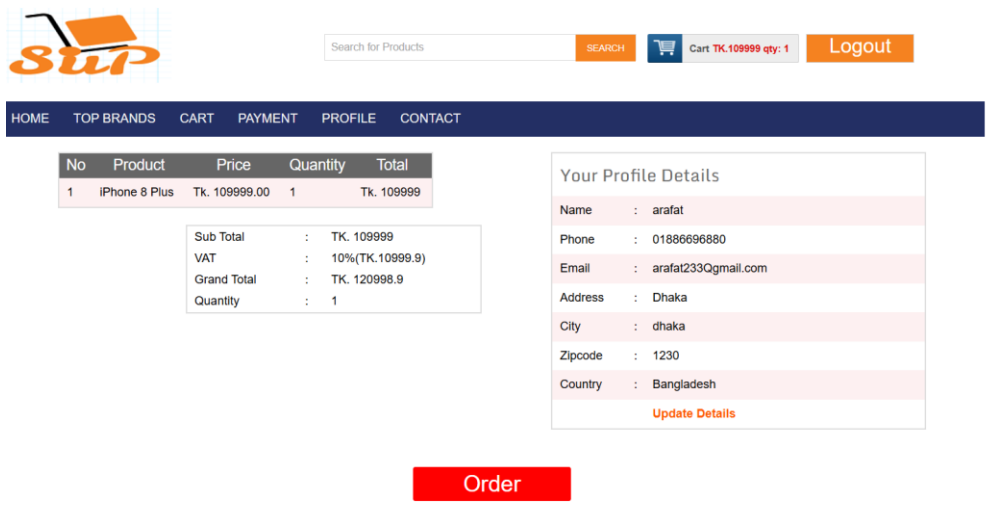
3. Click "Add to Cart"



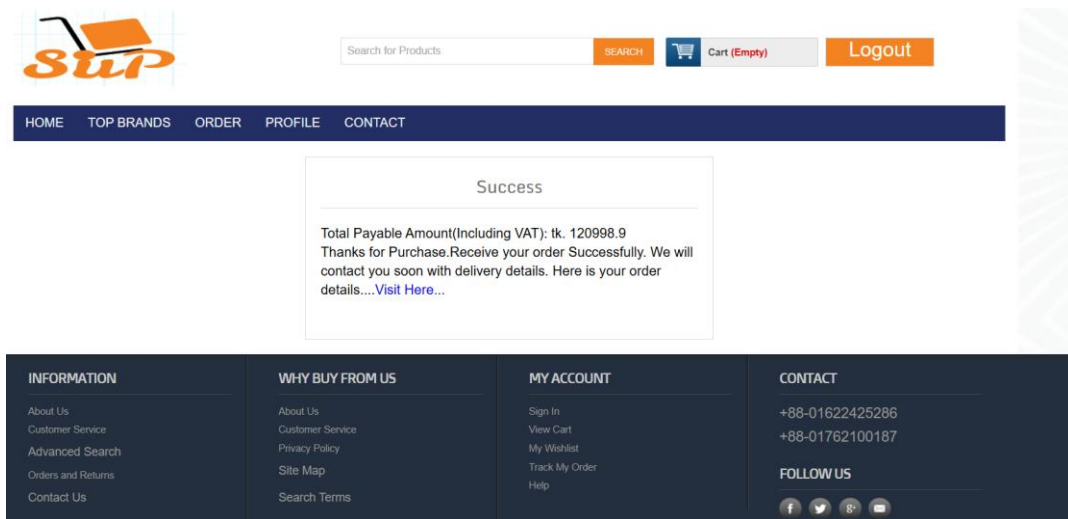
4. Go to cart page



5. Click checkout



6. Confirm order



7. Quality Assurance & Testing

7.1 Test Cases

ID	Test	Input	Expected	Actual	Status
TC-1	Login	valid data	success	success	Pass
TC-2	Cart	add item	added	added	Pass
TC-3	Order	confirm	saved	saved	Pass

7.2 RTM

FR Test Case

FR-1 TC-1

FR-5 TC-2

FR-7 TC-3

8. Individual Contribution

Yeasir Arafat

- Designed frontend UI
- Implemented product pages

Challenge: Responsive design issue

Solution: CSS media query

Learning: HTML, CSS, UI structure

Arif Hasan

- Developed backend logic
- Managed database

Challenge: SQL query error

Solution: Debugging and testing

Learning: PHP & database integration

Sayem Islam Tamim

- Software Testing & Quality Assurance
- Testing each feature step by step
- Checking database queries carefully
- Using debugging techniques (print, echo, error checking)
- Retesting after fixing bugs

Challenge Solution: Fixed Login Validation

Resolved Cart Issues

Improved Test Case Design

Corrected Database Queries

9. Conclusion & Future Work

9.1 Conclusion

The system successfully provides a digital solution for online shopping. It improves efficiency, accessibility, and user experience while simplifying business operations.

9.2 Future Enhancements

- Online payment (bKash, Nagad)
- Mobile application
- AI-based recommendation system
- Real-time order tracking

10. Appendices

Appendix A:

GitHub Repository Link : <https://github.com/arafat23617/Online-Shopping-System>

Appendix B: Setup Guide

1. Install XAMPP
2. Start Apache & MySQL
3. Import db_shop.sql
4. Move project to htdocs
5. Open localhost/project